

di-
vi-
fi-
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tion

×
??
×
??

First

$\gamma =$
0.272
 $t =$
2.18
0.087

×
×
 $r =$
0.376
 $P =$
58.8
 $p <$
0.001

Second

8.63/*VIX*
−13.7%
−33.8%
−21.2%

$t =$
0.86
??
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??
??
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??
??
??
??

0050.TW
TWII
(TAIEX)

Nine

in-
di-
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ual
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wan
stocks
and
one
ETF

??
U.S.
bench-
marks
Asia-
Pacific
mar-
kets

VIXTWN

$r_t =$
 $\ln(P_t/P_{t-1}) \times$
100
??

$\gamma_{GJR}t(\gamma)$

−0.31
−0.47
−0.52
−0.18
−0.21
−0.22

Notes:

γ_{GJR}
 $w =$
2000
 $t(\gamma)$
??

**GJR-
GARCH(1,1)**

$\sigma_t^2 = \omega + (\alpha + \gamma \cdot 1_{r_{t-1} < 0}) r_{t-1}^2 + \beta \sigma_{t-1}^2$

(1)

$\gamma >$
0

GARCH(1,1)

$\gamma =$
0

EWMA

(Ex-
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tially
Weighted
Mov-
ing
Av-
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age)

$$w_t = \min\left(\frac{K}{VIX_t}, 1\right)$$

(6)

$$\frac{K}{12/1.39} = \frac{8.63}{12/VIX}$$

$$\frac{Z}{16.2} = \frac{p}{0.001}$$

$$\frac{K}{12/1.39} = \frac{8.63}{43\%}$$

$$\frac{\times}{1}$$

??³
di-
tional
change

+32%
+69%
≈